Magic Square Sudoku
(The Golden Lo Shu Square)

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Abstract

Sudoku with the use of a 3-by-3 Magic Square exhibits symmetric properties which can be utilized to solve the puzzle with 16 clues.

						9		
					7			
3								
			2	7	6		8	
	4		9		1			
			4	3	8			
								7
				8				
		1						

Figure 1: Sudoku with 16 clues.

1 Introduction

Sudoku Magic Square is a study of the application of 3x3 Magic Squares to Sudoku. Sudoku Magic Square is a new variant of Sudoku where each 3x3 sub-grid forms at most a Magic Square and at least Latin Squares. The Magic Square will first be defined before applying it to Sudoku. The process of applying a Magic Square to Sudoku will highlight the properties of Sudoku Magic Squares.

1.1 Prerequisite

To solve the Sudoku of figure 1, the three permutations should be understood. The first permutation, σ takes an ordered 3x3 grid to a Magic Square. The ρ and κ permutations are cyclic permutations of the 3-groups horizontally and vertically respectively.

The following introductory branches were used for deeper understanding of a 16-clue Sudoku:

- topology
- group theory
- modulo arithmetic

The construction can be understood as a covering over a torus, or as a modulo-3 graph.

2 Construction

2.1 3-by-3 Grids

It is only necessary to start with a Magic Square with 1-2-3 ordering, and the use of the Magic Square graph for each permuted 3-group in every sub-grid.

The permutation for the Magic Square is:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 7 & 6 & 9 & 5 & 1 & 4 & 3 & 8 \end{pmatrix}$$

.

1	2	3
4	5	6
7	8	9

$\rightarrow$

2	7	6
9	5	1
4	3	8

2.2 9-by-9 Grids

Permutations between sub-grids horizontally across the Sudoku board:

$$\rho = (1\ 2\ 3)(4\ 5\ 6)(7\ 8\ 9)$$

.

2	7	6	3	8	4			
9	5	1	7	6	2			3
4	3	8	5	1	9			
5	1	9						
3	8	4			5			6
7	6	2						
		7			8			9

Figure 2: 9-by-9 Sudoku Magic Square composed of Semi-Magic and Magic Squares

Similarly, column permutations are identified vertically along the Sudoku board:

$$\kappa = (1\ 4\ 7)(2\ 5\ 8)(3\ 6\ 9)$$

.

2.3 Magic Square Sudoku

These permutations relate the Magic Square, semi-Magic Square and the Latin square properties to the Sudoku grid. The Golden Lo Shu Square is the complete transformation.

Figures 3 and 4 show symmetric properties horizontally and vertically but do not lend themselves to a unique solution because rows or columns without clues can be swapped in the final solution.

Going from figure 3 to figure 4, it is the first attempt to move clues to rows and columns in order to restrict swapping.

Finally, figure 4 to figure 5, $2 \to 7$ and $6 \to 8$. There is then only 1 row and 1 column with no clues.

No rows or columns can be swapped in the final solution.

Figure 5 completes the special case of Sudoku with 16 clues using three permutations and two swaps.

	1			2			3	
			2	7	6			
	4		9		1		6	
			4	3	8			
	7			8			9	

Figure 3: Sudoku with multiple solutions.

						9		
				2				
3								
			2	7	6			
	4		9		1		6	
			4	3	8			
								7
				8				
		1						

Figure 4: Sudoku with multiple solutions.

						9		
					7			
3								
			2	7	6		8	
	4		9		1			
			4	3	8			
								7
				8				
		1						

Figure 5: Sudoku with 16 clues.

The final solution is unique - there are no rows or columns in the final solution that can be swapped without moving the starting condition of clues. The final solution can be verified using the above stated permutations.

References

McGuire, Gary, Bastian Tugemann, and Gilles Civario (2013). "There is no 16-Clue Sudoku: Solving the Sudoku Minimum Number of Clues Problem via Hitting Set Enumeration". In: *arXiv preprint arXiv:1201.0749*. URL: https://arxiv.org/abs/1201.0749.

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